interaction history (e.g., routing based on prompt caching (Gu et al., 2025)). When $q(x)$ is small but positive on a non-negligible fraction of typical prompts, the mixture in (3) significantly complicates discrimination; nevertheless, a sound detection procedure must retain statistical power under such adversarial conditions.

Detector and objectives. A *detector* is a decision rule $\delta : \mathcal{D}_N \to \{0, 1\}$ whose output 1 means "reject H_0". Here, $\mathcal{D}_N$ denotes the data collected with N API queries and many local model queries. For a user-specified significance level $\alpha \in (0, 1)$, the detector must satisfy the type-I error constraint $\Pr_{H_0}\left[\delta(\mathcal{D}_N) = 1\right] \leq \alpha$. Subject to this guarantee, the verifier seeks to maximize the statistical power

$$\beta(q, \pi_{\text{alt}}) \;=\; \Pr_{H_1}\big[\delta(\mathcal{D}_N) = 1\big],$$

uniformly over a broad class of substitution policies (q, π_{alt}); to do so with as few costly API queries N as possible; and to employ prompts that are indistinguishable from ordinary user traffic, thereby preventing the provider from detecting and rerouting "suspicious-looking" inputs.

4 Method

In this section, we introduce our rank-based uniformity test in Section 4.1 and empirically identify the most discriminative score function to maximize the detection power in Section 4.2.

4.1 Rank-Based Uniformity Test (RUT)

To determine whether the target model π_{tgt} is identical to the reference model π_{ref}, we propose a *rank-based uniformity test*. This method assesses how typical the target model's responses are under the reference model's generation distribution.

Response score function. To perform a statistical test, we first define a scalar-valued *score function* $f : \mathcal{Y} \times \mathcal{X} \to \mathbb{R}$ that maps a model response and prompt to a real number. This function assigns a score to each output given the prompt, i.e.,

$$s = f(y, x), \quad \text{where } x \in \mathcal{X},\ y \in \mathcal{Y},\ s \in \mathbb{R}.$$

An ideal score function f^* should induce an *injective* mapping $y \mapsto f^*(y, x)$ for any fixed prompt $x \in \mathcal{X}$. Under this assumption, each distinct response corresponds to a unique score value, ensuring that the score distribution fully characterizes the model's outputs.

Uniformity as a test signal. For each prompt $x \in \mathcal{X}$, we sample a response $y_{\text{tgt}} \sim \pi_{\text{tgt}}(\cdot \mid x; \varphi)$ and compute its scalar score $s_{\text{tgt}} = f(y_{\text{tgt}}, x)$. To assess how typical this response is under the reference model, we evaluate its rank in the reference model's score distribution.

We define the cumulative distribution function (CDF) of the reference model's scores as:

$$F_{\pi_{\text{ref}}}(s \mid x) := \mathbb{P}_{y \sim \pi_{\text{ref}}(\cdot \mid x; \varphi)}\left[f(y, x) \leq s\right].$$

Since $f(y, x)$ takes values in a discrete set, $F_{\pi_{\text{ref}}}$ is a step function. To ensure the rank statistic is continuously distributed under the null hypothesis, we apply a *randomized quantile residual* (Dunn and Smyth, 1996) to extend the probability integral transform (David and Johnson, 1948) to discrete distributions. Specifically, we define the *rank statistic* as

$$r_{\text{tgt}} := F_{\pi_{\text{ref}}}(s_{\text{tgt}}^{-}) + U \cdot \mathbb{P}\left(f(y, x) = s_{\text{tgt}}\right), \quad U \sim \text{Uniform}[0, 1], \tag{4}$$

where $F_{\pi_{\text{ref}}}(s_{\text{tgt}}^{-}) := \mathbb{P}\left(f(y, x) < s_{\text{tgt}}\right)$ is the left-limit of the CDF at s_{tgt}, and $\mathbb{P}(f(y, x) = s_{\text{tgt}})$ is the probability mass at s_{tgt}. Under the null hypothesis $\pi_{\text{tgt}} = \pi_{\text{ref}}$, this rank statistic $r_{\text{tgt}} \in [0, 1]$ is uniformly distributed.

Conversely, suppose that $r_{\text{tgt}} \sim \text{Uniform}[0, 1]$ under the randomized quantile residual construction. Since the CDF $F_{\pi_{\text{ref}}}(\cdot \mid x)$ is stepwise and non-decreasing, a uniformly distributed r_{tgt} implies that the score s_{tgt} follows the same discrete distribution as s_{ref}. By injectivity of f, this further implies that $y_{\text{tgt}} \sim \pi_{\text{ref}}(\cdot \mid x; \varphi)$, and hence $\pi_{\text{tgt}} = \pi_{\text{ref}}$.

Thus, with an injective score function f, testing the uniformity of r_{tgt} as defined in (4) offers a valid signal for distinguishing π_{tgt} from π_{ref}.

Empirical approximation of $F_{\pi_{\text{ref}}}$. In practice, it is intractable to build the true CDF $F_{\pi_{\text{ref}}}(\cdot \mid x)$. Instead, we approximate it using an empirical CDF from m reference samples for each prompt.

Given a target response $y_i \sim \pi_{\text{tgt}}(\cdot \mid x_i; \theta)$ and reference responses $y_{ij} \sim \pi_{\text{ref}}(\cdot \mid x_i; \theta)$ for $j = 1, \ldots, m$, we compute the scalar scores

$$s_i := f(y_i, x_i), \quad s_{ij} := f(y_{ij}, x_i).$$

We then define the *randomized rank statistics* $r_i \in [0, 1]$ as

$$r_i = \frac{1}{m} \left(\sum_{j=1}^{m} \mathbf{1}\{s_i > s_{ij}\} + U_i \cdot \sum_{j=1}^{m} \mathbf{1}\{s_i = s_{ij}\} \right),$$

where $U_i \sim \text{Uniform}[0, 1]$ is an independent random variable to break ties uniformly, and ensure r_i is an unbiased estimator of r_{tgt} given the prompt x_i.

Discriminative score function via empirical selection. While an ideal *injective* score function would guarantee sensitivity to any behavioral difference between π_{tgt} and π_{ref}, constructing such a function for which we can calculate the CDF is generally infeasible in practice.

To ensure that our test remains practically effective, we instead require the score function to be *sufficiently discriminative*, in the sense that it induces distinct score distributions whenever $\pi_{\text{ref}} \neq \pi_{\text{tgt}}$. Formally, for fixed prompt $x \in \mathcal{X}$, let

$$S_{\pi_{\text{ref}}} := f(y, x) \text{ with } y \sim \pi_{\text{ref}}(\cdot \mid x; \varphi), \quad \text{and} \quad S_{\pi_{\text{tgt}}} := f(y, x) \text{ with } y \sim \pi_{\text{tgt}}(\cdot \mid x; \varphi).$$

We say that f is sufficiently discriminative if the distributions of $S_{\pi_{\text{ref}}}$ and $S_{\pi_{\text{tgt}}}$ differ whenever $\pi_{\text{ref}} \neq \pi_{\text{tgt}}$, i.e.,

$$\pi_{\text{ref}}(\cdot \mid x; \varphi) \neq \pi_{\text{tgt}}(\cdot \mid x; \varphi) \quad \Rightarrow \quad P_{S_{\pi_{\text{ref}}}} \neq P_{S_{\pi_{\text{tgt}}}}.$$

Under this condition, differences in response distributions are reflected in the score distributions, causing the ranks to deviate from uniformity.

Thus, we aim to find the most discriminative score function among several promising candidates through empirical experiments. In Section 4.2, we compare five candidate score functions—log-likelihood, token rank, log-rank, entropy, and the log-likelihood log-rank ratio (Su et al., 2023)—and find that log-rank is the most discriminative in practice for separating responses by π_{ref} and π_{tgt}, and therefore adopt it in our uniformity test.

Full test procedure. We now present the full RUT procedure.

Let $\{x_1, \ldots, x_n\} \subset \mathcal{X}$ be a set of prompts. For each prompt x_i, we sample one response from the target model,

$$y_i \sim \pi_{\text{tgt}}(\cdot \mid x_i; \theta),$$

and m responses from the reference model,

$$y_{ij} \sim \pi_{\text{ref}}(\cdot \mid x_i; \theta), \quad j = 1, \ldots, m.$$

We compute the log-rank scores

$$s_i := f(y_i, x_i), \quad s_{ij} := f(y_{ij}, x_i),$$

and the corresponding randomized rank statistics $\{r_i\}_{i=1}^{n}$.

We apply the Cramér–von Mises (CvM) test (Cramér, 1928) to assess the deviations between $\{r_i\}_{i=1}^{n}$ and $\text{Uniform}[0, 1]$. The test evaluates the null hypothesis

$$H_0 : r_i \sim \text{Uniform}[0, 1] \quad \text{for all } i.$$

The CvM test statistic is defined as

$$\omega^2 = \frac{1}{12n} + \sum_{i=1}^{n} \left(\frac{2i - 1}{2n} - r_{(i)} \right)^2,$$

where $r_{(1)} \leq r_{(2)} \leq \cdots \leq r_{(n)}$ are the ordered rank statistics.

To compute the p-value, we compare the observed statistic ω^2_{obs} to the distribution of the CvM statistic ω^2_{null} computed under the null hypothesis. The p-value is given by

$$p\text{-value} = \mathbb{P}_{H_0} \left[\omega^2_{\text{null}} \geq \omega^2_{\text{obs}} \right].$$

We reject H_0 and conclude that the target and reference models are different if p-value < 0.05.

4.2 Score Function Selection

The RUT requires a scalar score function $f(y, x)$. To identify a function that best captures distributional differences between models, we consider five candidate functions:

- **Log-likelihood:** $\log \pi_{\text{ref}}(y \mid x)$.
- **Token rank:** the average rank of response tokens in y, where a token's rank is its position in the vocabulary ordered by the π_{ref}'s next-token probabilities.
- **Log-rank:** the average of the logarithm of the token rank.
- **Entropy:** predictive entropy for y under $\pi_{\text{ref}}(x)$.
- **Log-likelihood log-rank ratio (LRR):** the ratio between log-likelihood and log-rank. (Su et al., 2023).

To identify the most discriminative score function, we conduct a Monte Carlo evaluation consisting of 500 independent trials. In each trial, we randomly select 10 prompts from the WildChat (Zhao et al., 2024) dataset and sample 50 completions per prompt from both π_{ref} and π_{tgt}, using a fixed temperature of 0.5 and a maximum length of 30 tokens. For each candidate score function, we compute the average AUROC (Bradley, 1997) across the 10 prompts for each trial, yielding a distribution of 500 AUROC scores per function. The full algorithm to calculate per score function average AUROC is included in Appendix A.1. Across different model comparisons, we find that **log-rank** consistently yields the most separable AUROC distribution from 0.5, indicating the strongest discriminative power. Figure 2 shows an example comparing Gemma-2-9b-it with its 4-bit quantized variant. Based on these results, we select log-rank as the scoring function for our uniformity test. Complete AUROC results are provided in Appendix A.2.

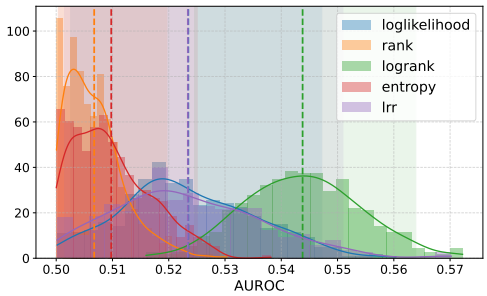

Figure 2: Distribution of AUROC scores for five candidate score functions across 500 trials comparing Gemma-2-9b-it and its 4-bit quantized variant. Log-rank achieves the most separable distribution from the random level 0.5, indicating superior power in distinguishing different models.

5 Experiments

In this section, we evaluate RUT across diverse model substitution scenarios, including quantization (Section 5.2), jailbreaks (Section 5.3), SFT (Section 5.4), full model replacement (Section 5.5), and real-world API providers (Section 5.6). Detection performance is compared against MMD and a KS baseline using statistical power AUC as the primary metric.

5.1 Experimental Setup

To evaluate detection performance under adversarial conditions, we simulate probabilistic substitution attacks where a fraction $q \in [0, 1]$ of API queries are routed to an alternative model (e.g., quantized or fine-tuned). For each value of q, we estimate the statistical power, defined as the probability of correctly rejecting the null hypothesis when substitution is present. We then summarize the resulting power–substitution rate curve using the area under the curve (AUC) over $q \in [0, 1]$. The AUC ranges from 0 to 1 and reflects the method's ability to maintain high statistical power across varying levels of substitution, serving as a measure of robustness to such attacks. Higher values indicate more reliable and consistent detection performance. Figure 1 shows an example comparing Gemma-2-9b-it and its 4-bit quantized variant.

Data. We use the WildChat dataset (Zhao et al., 2024), which contains real-world conversations between human users and ChatGPT. This dataset reflects authentic user behavior, ensuring the query distribution remains indistinguishable from typical API traffic—crucial for evasion-resistant auditing.